Roll No.:..........................

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mechanical Engineering

Applied Thermodynamics (ME-42101)

Mid Semester Examination, **Date: 14-03-2024**

Section – A, B & DD

Time: 10:00 AM – 12:00 AM **Session:** Jan –July 2024 **B.Tech.** (4th Sem.) **MM:** 30

General Instructions:

i. All questions are graded equally. **Attempt any five questions, and question one is compulsory.**

ii. Please note the serial number of the question before attempting it.

iii. All students bring their own steam table and calculator. **Sharing of steam tables and calculators is not permitted.**

iv. Students must make sure that nothing is written on their Steam table.

v. The invigilator is able to examine any of the student steam tables.

Q.1. Multiple choice questions and Answers. (Attempt all) **(0.5 X 12 = 6)**

A. The INCORRECT statement about the characteristics of critical point of a pure substance is that **(CO-1/ BL-2)**

(a) there is no constant temperature vaporization process
(b) it has point of inflection with zero slope
(c) the ice directly converts from solid phase to vapor phase
(d) saturated liquid and saturated vapor states are identical

B. The slope of which of the following curve on a P-T diagram for the substance that expands on freezing will be negative **(CO-4/ BL-2)**

(a) melting (b) vaporization (c) sublimation (d) none of these

C. The efficiency of superheat Rankine cycle is higher than that of simple Rankine cycle because **(CO-3/ BL-3)**

(a) The enthalpy of main steam is higher for superheat cycle
(b) The mean temperature of heat addition is higher for superheat cycle
(c) The temperature of steam in the condenser is high
(d) The quality of steam in the condenser is low

D. What is the critical Temperature of water? **(CO-1/ BL-2)**

(a) 273^0 (b) 656^0 (c) 374^0 (d) 100^0

E. At triple point for water, which of the following term is not equal to zero? **(CO-1/ BL-2)**

(a) Enthalpy (b) Entropy (c) Internal energy (d) None of these

F. Dryness fraction of steam is defined as **(CO-1/ BL-2)**

(a) Mass of water vapour in suspension/(mass of water vapour in suspension + mass of dry steam)
(b) Mass of dry steam/mass of water vapour in suspension
(c) Mass of dry steam/(mass of dry steam + mass of water vapour in suspension)
(d) Mass of water vapour in suspension/mass of dry steam

G. For a Rankine cycle, which of the following is true? **(CO-3/ BL-2)**

(a) A reversible constant pressure heating process happens in steam boiler
(b) Reversible adiabatic expansion of steam in turbine
(c) Reversible constant pressure heat rejection in condenser
(d) All of the mentioned

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H. What is the effect of reheat on steam quality? (CO-3/ BL-2)

(a) Increase (b) Decrease (c) Does not change

(d) Depends on several parameters

I. The efficiency of an ideal regenerative cycle is ____ the Carnot cycle efficiency. (CO-3/ BL-2)

(a) Greater than (b) Equal to (c) Less than (d) None of the mentioned

J. Why both reheating and regeneration is used together? (CO-3/ BL-2)

(a) The effect of reheat alone on efficiency is very small

(b) Regeneration has a marked effect on efficiency

(c) Both of the mentioned

(d) None of the mentioned

K. For a given boiler and condenser pressure and mass flow rate of the steam, which of the following will decrease the specific work output of the Rankine cycle? (CO-1/ BL-2)

(a) Reheat (b) Superheating (c) Regeneration (d) None of the these

L. Define the term deaerator. Why is it used? (CO-3/ BL-2)

Q.2. A vessel having a capacity of 0.05 m^3 contains a mixture of saturated water and saturated steam at a temperature of 245°C. The mass of the liquid present is 10 kg. Find (i) The pressure, (ii) The mass, (iii) The specific volume, (iv) The specific enthalpy, (v) The specific entropy, and (vi) The specific internal energy. **(CO-1/ BL-2)**

Q.3. Calculate the internal energy per kg of superheated steam at a pressure of 10 bar and a temperature of 300°C. Also find the change of internal energy if this steam is expanded to 1.4 bar and dryness fraction 0.8. **(CO-1/ BL-2)**

Q.4. A Rankine cycle operates between pressures of 80 bar and 0.1 bar. The maximum cycle temperature is 600°C. If the steam turbine and condensate pump efficiencies are 0.9 and 0.8 respectively, calculate the specific work and thermal efficiency. **(CO-3/ BL-3)**

Q.5. In a 15 MW steam power plant operating on ideal reheat cycle, steam enters the H.P. turbine at 150 bar and 600°C. The condenser is maintained at a pressure of 0.1 bar. If the moisture content at the exit of the L.P. turbine is 10.4%, determine: (i) Reheat pressure; (ii) Thermal efficiency; (iii) Specific steam consumption; and (iv) Rate of pump work in kW. Assume steam to be reheated to the initial temperature. **(CO-3/ BL-3)**

Q.6. In a single-heater regenerative cycle the steam enters the turbine at 30 bar, 400°C and the exhaust pressure is 0.10 bar. The feed water heater is a direct contact type or open type which operates at 5 bar. Find the efficiency and the steam rate of the cycle. Pump work may be neglected. **(CO-3/ BL-3)**

****************************BEST WISHES***************************

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mechanical Engineering

Mid Semester Examination Jan- June:2024

Course Name: Industrial Engineering and Management **Full Marks-30**

Course Code: ME42105 **Total Time: 120 min**

Instructions: (i) Answer all question

(ii) All question carries equal marks

1. For product demand and forecasting values for various periods are given in the table

Period	Demand	Forecast value
A	110	96
B	124	127
C	119	134
D	134	112

Find the sum of forecast error and mean absolute deviation.

2. (a) What is forecasting in industrial engineering? and its advantages

   (b) The actual demand for a product in company is 79 units. The previous forecast and exponential smoothing factor are 84 units and 0.25 respectively. What will be the forecast for next period.

3. (a) what is the basic theory of linear programming?

   (b) A firm manufacturing two product A and B. on which profit earn per pieces are Rs.3 and Rs.4 respectively. Each product is processed on 2 machines M_1 and M_2. A requires 1 min of processing time on machine M_1 and 2 min on M_2. While B requires 1 min on M_1 and 1 min on M_2. M_1 is not available for more than 7 hrs 30 min while M_2 is available for 10 hrs only. Determine the number of units of product A and B should be produces to get maximum profit.

4. (a) What are the different types of forecasting techniques?

   (b) Explain the different types of errors in forecasting in industrial engineering.

5. What do you understand by Job, Shop and Mass production

National Institute of Technology Patna

Department of Mechanical Engineering

Mid Semester Examination

Subject: Kinematics of Machinery — Subject Code: ME42104

Date: 19/03/2024 — Time: 2 hr

Answer Any Three Questions [10×3 =30]

1. **a.)** Determine degree-of-freedom of the planer mechanism as shown in Fig.1 and Fig.2.
   b.) Explain the terms with appropriate example: i.) Lower pair, ii.) Higher pair, iii.) Kinematic chain and Mechanism. **[4+6=10]**

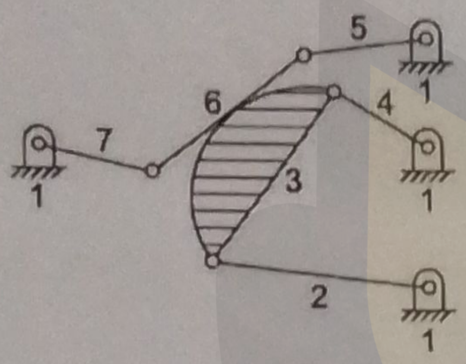

Fig. 1

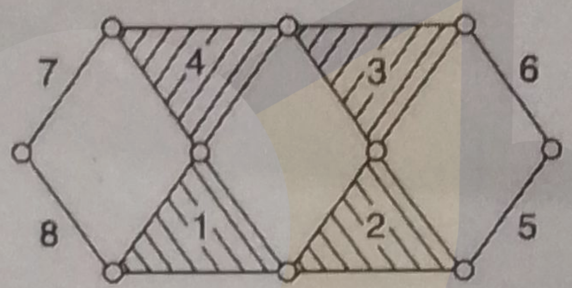

Fig. 2

2. Discuss the inversions of single and double slider four bar mechanism and their applications. **[10]**

3. A quick return mechanism of the crank and slotted lever type shaping machine is shown in Fig 3. The dimensions of the various links are as follows : O_1O_2 = 800 mm ; O_1B = 300 mm ; O_2D = 1300 mm ; DR = 400 mm. The crank O_1B makes an angle of 45° with the vertical and rotates at 40 r.p.m. in the counter clockwise direction. Find : i.) velocity of the ram R, or the velocity of the cutting tool, and ii.) angular velocity of link O_2D. **[10]**

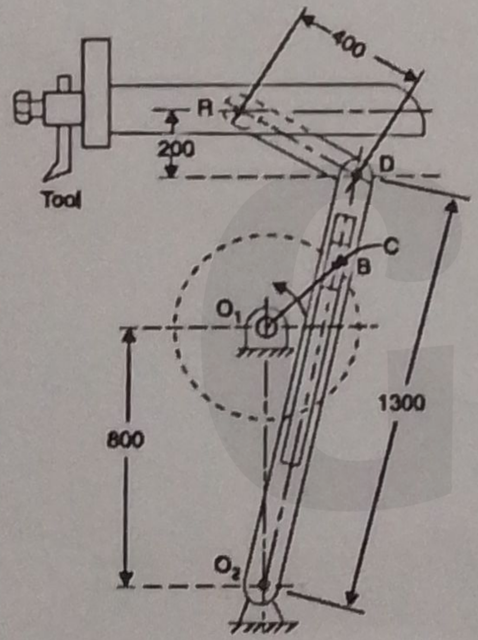

Fig. 3

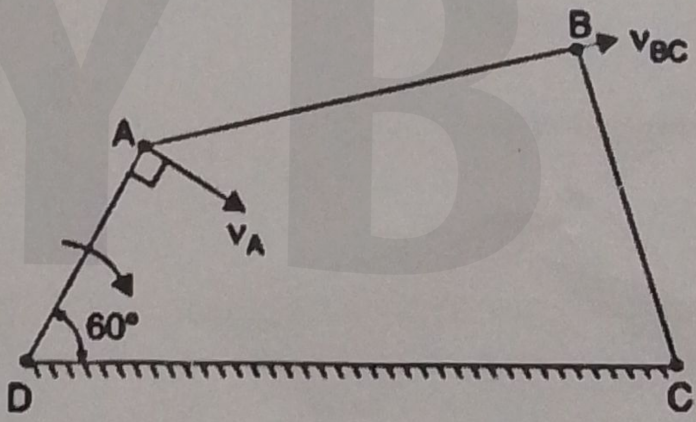

Fig. 4

4. A four bar mechanism has the following dimensions :
   DA = 300 mm ; CB = AB = 360 mm ; DC = 600 mm. The link DC is fixed and the angle ADC is 60° as shown in Fig. 4. The driving link DA rotates uniformly at a speed of 100 r.p.m. clockwise and the constant driving torque has the magnitude of 50 N-m. Determine the velocity of the point B and angular velocity of the driven link CB. Also find the actual mechanical advantage and the resisting torque if the efficiency of the mechanism is 70%. **[10]**

National Institute of Technology Patna
Mid Semester Exam Jan-June 2024
Manufacturing Process-II (ME42103)
B.Tech. IV Sem (Section – B)

Time: 2 Hours **M. Marks: 30**

Note: Attempt all questions.

1. What is high energy rate forming processes? Explain the process of forming by using explosives with a schematic diagram. [4]

2. What is sheet metal working process? Describe any three shearing operations with diagrams. [4]

3. Describe various stages of powder metallurgy process. [4]

4. A strip with initial dimension of $24 \times 24 \times 150$ mm^3 is forged between the two flat dies to the final size of $6 \times 96 \times 150$ mm^3. If the coefficient of friction is 0.05. Determine the maximum forging force. Take average yield strength of material as 8 MPa. [8]

5. Analyze the tube drawing process with a conical plug and establish the relation for drawing stress. [10]

OR

Analyze the wire drawing process and establish the relation for drawing stress. [10]

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mechanical Engineering

MID SEMESTER EXAMINATION, January - June 2024

B.Tech and DD in Mechanical Engineering | **Semester – IV** | **Section - B**

Course Name: Fluid Mechanics and Machinery | **Course Code: ME42102**

Time: 2 hours | **Full Marks: 30**

Instruction:

1. *Attempt any three questions.*
2. *Assume any suitable data, if necessary.*
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to question are mentioned.*

1. (a) Define dynamics viscosity and kinematic viscosity of fluid.

(b) Define compressible and incompressible fluid flow based on Mach number (Ma) in details.

(c) A uniform film of oil 0.13 mm thick separates two discs, each of 200 mm diameter, mounted coaxially. Ignoring the edge effects, calculate the torque necessary to rotate other at a speed of 7 rev/s, if the oil has a viscosity of 0.14 Pas.

3+3+4 CO1 BL1/2/3

2. (a) States the different condition of stability of submerged body with clear figures.

(b) Derive the equation of pressure difference using Micro-manometer with clear figure.

(c) An inclined tube manometer measure the gauge pressure P_s of a system S (Fig. 1). The reservoir and tube diameter of the manometer are 50 mm and 5 mm respectively. The inclination angle of the tube is 30°. What will be the percentage error in measuring P_s if the reservoir deflection is neglected?

3+3+4 CO1 BL1/2/3

3. (a) Prove that the relation of floating body for stable condition $GM = \frac{I_{yy}}{V} - BG$, indicated terms are the conventional parametric abbreviation.

(b) A cube of side 'a' floats with one of its axes vertical in a liquid of specific gravity S_L (Fig. 2). If the specific gravity of the cube material is S_c, find the value of S_L/S_c for the metacentric height to be zero.

5+5 CO1 BL1/3/3

4. (a) What is the effect of temperature on viscosity of liquid and gases.

(b) Derive the equation of pressure variation in compressible fluid in under Isothermal and Non-isothermal condition.

(c) A two dimensional gate having cross-section of a quarter cylinder is pivoted (hinged, Fig. 3) at the bottom O and is in static equilibrium when subjected to differential fluid of specific weight γ_1 and γ_2. Assuming the gate weightless, calculate the relationship between γ_1 and γ_2.

2+3+5 CO1 BL1/2/3

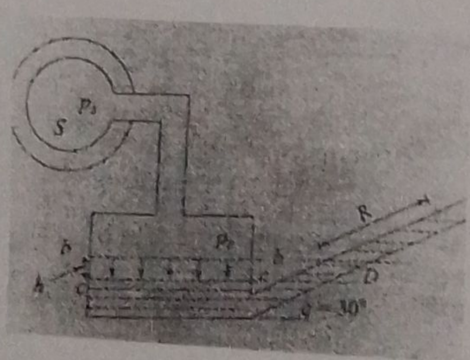

Fig. 1

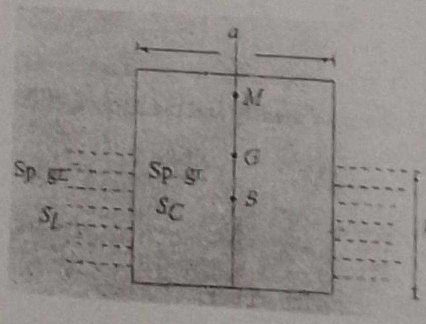

Fig. 2

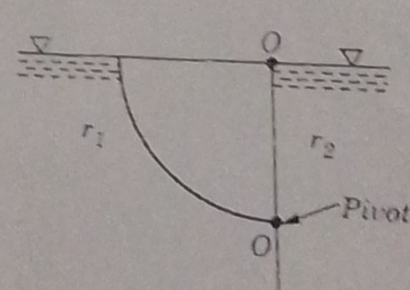

Fig. 3

National Institute of Technology, Patna
Department of Mechanical Engineering

Mid semester examination (Jan-June 2024)

Time: 2 hrs. | Sem. IV th (UG), Course: Introduction to Robotics (ME42106) | Full Marks: 30

*Answer **all** questions. The marks related to the questions are mentioned on the right hand side margin*

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1. (a) Fill in the blanks with appropriate words: 4

i. Robot with fixed base can divided in to .... types

ii. Rotation matrix about y axis is ....

iii. .... Robot has two linear and one rotational movement

iv. The term robotics was introduced by .... In the year ....

(b) i. Define degree of freedom. 2

ii. What is the role of D-H notation? 2

iii. Describe the types of joints used in robots with simple sketches. 2

2. (a) Enumerate the difference between open loop and closed loop control system. 5

(b) Determine the revolution matrix for a rotation of 45^0 about y-axis followed by a rotation of 120^0 about z-axis, and a final rotation of 90^0 about x-axis. 5

3. (a) Briefly describe the SCARA robots. 5

(b) Discuss about the salient features of servo and stepper motor with limitations. 5

======================== **BEST WISHES** ========================